Homework 1: Direction of Arrival estimation

Prof. Andrea Virgilio Monti-Guarnieri
Dr. Marco Manzoni

A.Y. 2024/2025

Instructions

- The output of the homework is a set of MATLAB files and a report (Word, LaTeX) containing the answers to the proposed questions, the required figures, and a brief explanation. The MATLAB files must be error-free and should not require user input.
- The homework is divided in steps with increasing difficulty. Partial solving of the homework is allowed.
- You can work in groups to share ideas, doubts, and possible solutions. However, The final report is **individual**, meaning that each student should generate their own figures and write their own report. You will be asked questions about the report during the oral discussion, so be ready to defend your work.
- The text doesn't ask to plot specific figures or show some specific equations used. It is up to the student to decide which figure will better convey the information to the reader and to produce it.
- For each homework there are lessons dedicated to classwork. During these hours, you can work and ask the instructor questions. No questions are allowed outside these lessons.
- The text defines requirements and parameters. Missing information -if any- are a free choice of the engineer.
- The deadline: **08/12/2024** (included).
- Compress all files in a ZIP package and call it `name_surname_HW1_XXXXXXXX.zip`, where XXXXXXXX is the ID number of the student (e.g. `marco_manzoni_HW1_10455829.zip`)
- The ZIP must be sent by email at marco.manzoni@polimi.it and in Carbon copy (Cc) to andrea.montiguarnieri@polimi.it
- The oral discussion of the homework will be scheduled with a Doodle where you can reserve a time slot, usually between January and the end of February. At the oral, you will discuss both the homework. A theoretical question on a generic topic of the course and/or a quiz may be asked to assess your preparation.

1 System geometry

In this homework, you will try to estimate the angular position of a target using a Uniform Linear Array (ULA). The 2D geometry is depicted in Figure 1. The ULA is composed of N of antennas. The figure

shows an array with nine elements just for reference. All the elements are spaced by dx meters and are dislocated along the x direction. A single target is present in the scene, and it is depicted as a yellow star.

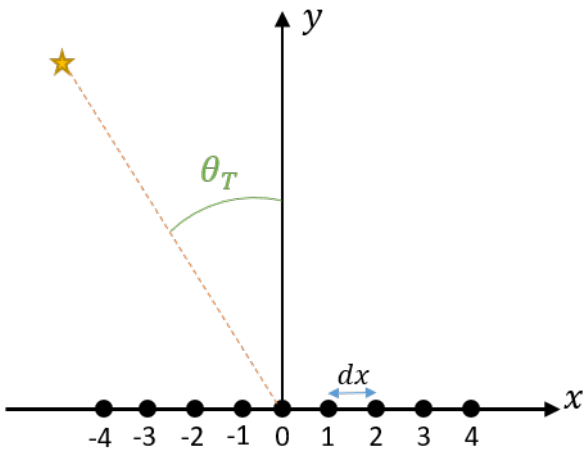


Figure 1: System geometry: nine elements are placed on the x axis with a reciprocal distance of dx. A target is represented as a yellow star, and it is located at an angular position θ_T

Each antenna transmits a sinusoidal signal $g(t) = e^{j2\pi f_0 t}$ with a frequency $f_0 = 77$ GHz and receives the echo back. This scenario, in which each antenna transmits a signal and only that antenna records the echo, is called *monostatic* configuration.

2 Direction of Arrival estimation (difficulty: easy)

In this section of the homework, you will design and simulate a ULA able to detect the angular position θ_T of the target in Figure 1 with a resolution at the boresight ($\theta = 0$) of 2 degrees. You will be guided through the design of the ULA in the following sections.

2.1 System model

Given the system geometry and the previously described transmitted signal, write the model of the signal received by the generic n^{th} antenna. Assume that the signal is perfectly demodulated at the receiver.

$$s_{RX}(n) = ? \tag{1}$$

Now, try to linearize the phase term of the received signal with respect to x (the antenna position) around the central element. Alternatively, you can use the plane wave approximation, leading to the same result. What do you find if you read the received signal along the elements? Can you recognize this signal?

Hints:

- The difference between the transmitted and received signal by each antenna is purely a delay that depends on the antenna-to-target distance.
- The time delay can be easily converted into a distance. Assume the speed of light in a vacuum.
- Demodulating means removing the carrier or multiplying the received signal by $e^{-j2\pi f_0 t}$.
- Assume that the complex reflectivity of the target is constant.
- Linearization of a generic function $f(x)$ around zero means making it *linear* with respect to x:
$$f(x) \approx f(0) + f'(x)|_{x=0}\Delta x$$
- In your scenario, look carefully at $f'(x)|_{x=0}$, what is that?
- Remember that the total length of the array is finite. Include this information in the signal model.

2.2 Array design and DoA estimation

Once you understand what kind of signal your antennas are capturing, reply to these three questions:

1. What should be the spacing between antennas?
2. Can you propose a way to detect the direction of arrival?
3. Can you design the total array length to obtain the desired finest resolution of 2 degrees?

Hints:

- To determine the spacing between elements, remember the condition to avoid aliasing when sampling a signal.
- Look again at the phase of the received signal; where is the angle θ_T?
- The angular resolution is the minimum angular distance at which two targets must be placed to distinguish them in the DoA estimation. For a cardinal sine function, the resolution is the distance from the peak to the first zero.
- To find the resolution of a generic array of length L, perform a sensitivity analysis: if I place another target at $\theta + \Delta\theta$, how much does the spatial frequency on the array change?
$$\Delta f_x = \frac{df_x(\theta)}{d\theta}\Delta\theta$$

After this section, you should have completed the design of the array, having defined its length, number of elements, their spacing, and the method to estimate the DoA. In the report, write all the reasoning and equations you used to derive the system's parameters.

3 MATLAB implementation (difficulty: easy)

Now that the design is ready, it is time to implement it in MATLAB. You must define a vector containing the antennas' spatial position, the target's location in the 2D space, and a vector for the received signal. Remember to respect the far-field condition when placing the target.

Then, try to use your algorithm to estimate the target's angular position. Here you have a few questions to which you should provide an answer:

- Compare the estimated DoA with the true angular position of the target. Is it the same?
- Is the resolution respected?
- Repeat the simulation with more than one target in the scene. What happens?
- What happens if you increase the spacing between antennas (dx)?

4 2D position estimation (difficulty: medium)

4.1 System model

In the previous section, the position estimation is just angular, meaning that the system is not able to estimate the distance between the array and the target (in radar jargon, this distance is called *range*). By changing the transmitted signal, it is possible to estimate the 2D position of the target (i.e., both in range and angle).

The transmitted signal in the time domain is now a cardinal sine function with a bandwidth of 1 GHz. Such a signal is modulated around a central frequency of 77 GHz as the previous one. The model of the transmitted signal by each antenna is then:

$$g(t) = \text{sinc}(Bt)e^{j2\pi f_0 t} \tag{2}$$

Now reply to these questions and write the reasoning and the equation that led you to the answers on the report.

1. What is the time and space resolution of such a signal?
2. What is the expression of the received and demodulated signal at each antenna given a generic target in the scene? The signal is no longer monodimensional (where the only independent variable was the antenna position), but it is bi-dimensional. The two variables are the time t (or equivalently, the range r, related to the time by $t = 2r/c$) and the antenna position.

4.2 MATLAB implementation

Implement the system just described in MATLAB. Assume that the system parameters are the same as the ones derived in the previous section of this homework. The received signal is now a matrix and not a vector. In each column, you insert the echo received by each antenna (a delayed version of the transmitted one). Try to perform range and DoA estimation.

5 MIMO array (difficulty: medium)

In real-world systems, arrays are frequently implemented using Multiple-Input Multiple-Output (MIMO) radar technology. In the scenario previously described, the N antennas both transmit and receive signals, a configuration known as *monostatic*. In contrast, a MIMO system consists of N_{Tx} transmitting antennas and N_{Rx} receiving antennas.

It can be shown that if you have two antennas, one transmitting and one receiving, the system behaves equivalently to a monostatic antenna with its effective phase center located at the midpoint between the transmitter and receiver. This effective phase center is referred to as a *virtual element*.

Clearly, if we have only one transmitting and one receiving antenna, we are using two physical antennas to create a single equivalent phase center, which is inefficient. With one transmitting antenna and N_{Rx} receiving antennas, we obtain N_{Rx} virtual elements using $N_{Rx} + 1$ physical antennas, which still isn't an optimal solution.

However, by using N_{Tx} transmitters and N_{Rx} receivers, we can generate a virtual array with $N_{Tx}N_{Rx}$ elements, using only $N_{Tx} + N_{Rx}$ physical antennas. For example, if $N_{Tx} = 2$ and $N_{Rx} = 4$, we generate a virtual array of 8 elements with just 6 physical antennas.

In Section 2, you have already defined the dimensions of the array. Now, imagine this as the virtual array. From here, reverse the process to determine the dimensions of the transmitting and receiving arrays. Once you have designed the array, simulate the system in MATLAB for the 1D case, as you did in Section 3. Be sure to simulate the actual MIMO system, not just the virtual array, or the simulation will be identical to the one in Section 3.

Hints:

- Remember that the product between N_{Tx} and N_{Rx} must be equal to the number of elements found in Section 2.
- Think deeply about the spacing between the transmitting and receiving elements. They must be placed in such a way that the resulting virtual array is spaced exactly as the one designed in Section 2. The only necessary information to design the transmitting and receiving array is that the virtual element is located in the midpoint between the TX antenna and the RX one.

6 Orthogonal waveform (difficulty: hard)

In Section 4 you have simulated a monostatic system in which each antenna transmits a cardinal sine function and receives the echo back. This system will work in a real scenario, only if each antenna transmits in Time Division Multiplexing (TDM), meaning that the transmission and receptions are consecutive and not simultaneous. If this is not the case, there is mutual interference between the signals, and the DoA estimation is not feasible.

Another approach is to transmit simultaneously orthogonal waveforms. Two waveforms $s_1(t)$ and $s_2(t)$ are said to be orthogonal if their cross-correlation is zero:

$$r(t) = s_1(t) * s_2^*(-t) = \int s_1(\tau)s_2^*(\tau + t)d\tau = 0 \tag{3}$$

The signal received by the kth antenna can be written as:

$$s_{Rx}^{k}(t) = \sum_{n=1}^{N} s_n(t - \tau_n) e^{-j2\pi f_0 \tau_n} \tag{4}$$

where $s_n(t)$ is the signal transmitted by the $n-th$ antenna and τ_n is the delay induced by the propagation from the $n-th$ antenna to the target and back. The cross-correlation (range compression in radar jargon) is performed at each receiving antenna. In particular, at the $k-th$ antenna, if we want to range-compress the signal transmitted by the first antenna, we can match-filter using the first signal transmitted:

$$s_{Rx}^{k}(t) * s_1^*(-t) = \sum_{n=1}^{N} \int s_n(\tau - \tau_n) e^{-j2\pi f_0 \tau_n} s_1^*(\tau + t) d\tau \tag{5}$$

The summation can be split into two components, namely the one for which $n = 1$ and the one for which $n \neq 1$:

$$\int s_1(\tau - \tau_1) e^{-j2\pi f_0 \tau_1} s_1^*(\tau + t) d\tau + \sum_{n \neq 1} \int s_n(\tau - \tau_n) e^{-j2\pi f_0 \tau_n} s_1^*(\tau + t) d\tau \tag{6}$$

In virtue of Equation 3, the second one should be zero to avoid any interference between signals. The only remaining one is the component for which $n = 1$, which is the Impulse Response Function after matched filtering.

Now implement this kind of system. Consider possible waveforms to be transmitted, generate them, and perform the range compression to separate their contribution at the receiver.

Hints:

- The orthogonality condition can also be relaxed allowing for $r(t) \approx 0$ and not $r(t) = 0$. What kind of signal do you know with this property?